

JULIEN CHIQUET

CURRICULUM VITÆ

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TABLE OF CONTENTS

1	Curriculum Vitæ	2
1.1	Julien Chiquet, Researcher in Statistics, Habilitation	2
1.2	Main responsibilities	2
1.3	Summary of activities	2
1.4	Professional experience	2
1.5	Education	2
2	Scientific activities	3
2.1	Research enhancement	3
2.1.1	Team management	3
2.1.2	Scientific committees	3
2.1.3	Editorial activities	3
2.1.4	Scientific events	3
2.1.5	Important workgroups	3
2.2	Participation to research grants	4
2.2.1	Ongoing Projects	4
2.2.2	Past	4
2.3	Students	6
2.3.1	PhD and Post-doc - Current	6
2.3.2	PhD and Post-doc - Alumni	8
2.3.3	Masters - Alumni	10
2.4	Teaching activities	12
3	Scientific productions	13
3.1	Papers	13
3.1.1	Preprints	13
3.1.2	Journal papers	13
3.1.3	Conference papers (international)	16
3.1.4	Book chapters	17
3.1.5	Thesis	17
3.2	Talks	17
3.2.1	Invited talks	17
3.2.2	Research Schools and Tutorials	18
3.3	Software	18

CURRICULUM VITÆ

JULIEN CHIQUET, RESEARCHER IN STATISTICS, HABILITATION

Statistical Learning, Life Science, Reproducible Research

Sparse methods and regularization · Multivariate analysis · Latent variable models · Optimization and algorithms · Ecology, environment · Omics data



Born July 26, 1980
Married, father of 3 children

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🌐 <https://jchiquet.github.io>
🐙 <https://github.com/jchiquet>

Professional address

UMR MIA Paris-Saclay
Campus Agro Paris Saclay
22, place de l'agronomie
91120 Palaiseau, France

MAIN RESPONSIBILITIES

- Head of UMR MIA Paris-Saclay (since 2022)
- Creator and Chief Editor of *Computo* (since 2022).

SUMMARY OF ACTIVITIES

<i>Production</i>	48 journal papers, 6 preprints, 21 conference papers, 6 book chapters, 18 maintained software packages.
<i>Students</i>	4 ongoing PhD, 11 alumni, 24 Master.
<i>Grants</i>	3 ongoing, 23 past.
<i>Juries</i>	26 PhD + 5 HDR as reviewer, 12 PhD + 2 HDR as examiner.
<i>Teaching</i>	1824 undergraduate/MSc hours + ≈ 100 hours for continuing education courses.

PROFESSIONAL EXPERIENCE

since 2020	SENIOR RESEARCHER INRAE UMR MIA Paris-Saclay, AgroParisTech/INRAE Applied Mathematics and Computer Science
2020 – 2024	PART-TIME LECTURER X – École Polytechnique
2016–2019	FIRST CLASS RESEARCHER INRA UMR 518 MIA Paris, AgroParisTech/INRAE
2007 – 2015	RESEARCH AND TEACHING ASSISTANT, then ASSISTANT PROFESSOR (26 ^e section) UMR 8071 Statistique & Génome, Université d'Évry
2003 – 2007	PHD STUDENT French Nuclear Agency (CEA) Saclay

EDUCATION

2015	HABILITATION IN MATHEMATICS
<i>Title</i>	<i>Contributions to sparse methods for complex data analysis</i>
<i>Reviewers</i>	A. d'Aspremont (ENS), A. Dalalyan (ENSAE), J.-P. Vert (Mines ParisTech)
2003–2007	PHD IN APPLIED MATHEMATICS
<i>Title</i>	<i>Modeling and Estimating degradation processes with application in reliability</i>
<i>Supervisor</i>	French Nuclear Agency (CEA), Saclay Nikolaos Limnios (Université de Technologie de Compiègne)
2003	M.S. IN COMPUTATIONAL SCIENCE AND STATISTICAL LEARNING, Université de Technologie de Compiègne
2003	COMPUTER ENGINEERING SCHOOL, Université de Technologie de Compiègne

SCIENTIFIC ACTIVITIES

RESEARCH ENHANCEMENT

TEAM MANAGEMENT

	JOINT RESEARCH UNIT MIA PARIS-SACLAY , AGROPARISTECH/INRAE
<i>Since 2022</i>	Director of the lab
<i>2020–2021</i>	Deputy Director of the lab ; Team leader of team “SOLsTIS”

SCIENTIFIC COMMITTEES

	COUNCILS
<i>since 2020</i>	Member of the Steering Committee of Digit-Bio INRAE Metaprogramme
<i>since 2020</i>	Nominated member of the Scientific Council of the INRAE Animal Genetics division
<i>2016–2021</i>	Elected member of the Council of the French Statistical Society (2018–2021) and of the Scientific Council of the INRAE Math-Info division (2016–2020)
	BOARD OF RECRUITMENT
<i>Status</i>	1 Prof., 11 Assoc. Prof., 9 Junior Researchers, 6 Research Engineers
	PHD & HDR JURIES
<i>PhD</i>	26 jurys as Reviewer, 12 as Examiner.
<i>HDR</i>	5 jurys as Reviewer, 2 as Examiner.
	PHD FOLLOW-UP
<i>since 2014</i>	19 PhD students

EDITORIAL ACTIVITIES

	RESPONSIBILITIES
<i>since 2021</i>	Creator and Chief Editor of <i>Computo</i>
<i>2019 – 2022</i>	Associate Editor of the <i>Journal of Computational and Graphical Statistics</i>
<i>2018 – 2021</i>	Leader of the publication unit of the French Statistical Society
<i>Reviewer</i>	PAPER REPORTS
<i>Journal</i>	JMLR, JSS, JRSS-B, JRSS-C, Scandinavian, Biometrics, Biometrika, EJS, CSDA, JCGS, Stat. & Comp., Bioinformatics, BMC Bioinformatics, Plos Comp. Bio., ...
<i>Conference</i>	ECCB 2026, NIPS 2012–2017, ICML 2015, 2018, CAp 2025, ...

SCIENTIFIC EVENTS

	LEADER OF ORGANIZING COMMITTEES
<i>Conference</i>	RencontresR’21 , SMPGD (2019–2022), SatRDay’19
	PARTICIPATION IN SCIENTIFIC OR ORGANIZING COMMITTEES
<i>Conference</i>	RR 18/21/26, JDS 20/21, EcoStat 21, StatLearn’14, JFRB’14, IWAP 2008, ...
	RESEARCH SCHOOLS AND TUTORIALS
<i>Lecturer</i>	StatXP’19/20, Surf64’17/18, SPS’16, BigOptim’15, ... (see §3.2.2)

IMPORTANT WORKGROUPS

<i>since 2021</i>	DIGIT-BIO IA CELL
<i>Involvement</i>	Co-leader: webinar cycle to build a shared AI vocabulary, raise awareness of AI issues in life sciences, and identify biological questions for AI development.
<i>2017 – 2022</i>	WORKGROUP STATE OF THE R (FUNDING ≈ 5000€/YEAR)
<i>Involvement</i>	Co-founder and Co-leader, deepening know-how and exchanging around the latest innovations of R (annual bootcamp + monthly meeting, 20/30 people).
<i>2009 – 2022</i>	INRAE METHODOLOGICAL WORKGROUP NETBIO (FUNDING ≈ 5000€/YEAR)
<i>Involvement</i>	Co-leader between 2012 and 2022, network reconstruction methods in biology (annual meeting, 50 people; funded internships).

ONGOING PROJECTS

2023–2027	DISCERN – DISCOVERING THE CAUSES OF THREE POORLY UNDERSTOOD CANCERS IN EUROPE – https://discern.iarc.who.int/
<i>Partners</i>	20 partners
<i>Involvement</i>	Team leader, 250,000 € for MIA Paris-Saclay
<i>Description</i>	DISCERN investigates the causes of three poorly understood cancers (renal, pancreatic, colorectal) and the geographical disparities in their incidence across Europe, notably the higher rates observed in Central and Eastern Europe. The 20-institution consortium, coordinated by IARC, combines epidemiological, exposomic and multi-omics approaches within the Horizon Europe Mission on Cancer. I co-lead the methodological work package with Imperial College.
<i>Support</i>	Horizon Europe
2024–2027	COBREEDING – https://www.pepr-agroecinum.fr/les-projets-finances/socio-ecosysteme/projets-cibles/cobreeding
<i>Partners</i>	GQE Le Moulon, HeuDiaSyC (UTC)
<i>Description</i>	CoBreeding develops statistical and deep learning methods to predict the performance of maize hybrids from pools of parental lines, combining mixed models from quantitative genetics with neural network architectures, and transferring information between US and European maize line populations. Funds the PhD of François Victor.
<i>Support</i>	PEPR Agroécologie Numérique
2024–2027	HEPATOTWIN – https://digitbio.hub.inrae.fr/rubriques-verticales2/nos-actions/projets-phares/projet-phare-hepato-twin-2024-2028
<i>Partners</i>	INRAE (Toxalim, MIA-PS, NumeCan, UNH, TBI)
<i>Description</i>	HEPATOTWIN gathers INRAE units in toxicology, applied mathematics and nutrition to develop modelling approaches for liver toxicology.
<i>Support</i>	Metaprogramm DIGIT-BIO (Meta-omics and microbial ecosystems, INRAE)

PAST

2020–2025	EXPANSE: EXPOSOME POWERED TOOLS FOR HEALTHY LIVING IN URBAN SETTINGS – https://expanseproject.eu/
<i>Partners</i>	Imperial College, Utrecht, Columbia, Inserm, IARC + 15 others
<i>Description</i>	EXPANSE studies how the urban exposome – environmental exposures specific to city living – affects cardiometabolic and pulmonary disease, Europe’s leading disease burden. It combines exposome and health data from over 55 million Europeans, multi-omics data from 2 million individuals, and personalized exposure assessment for 5,000 participants across five urban regions.
<i>Support</i>	Horizon H2020
2020–2024	G2WAS – GRAPE GENES FOR WATER SCARCITY – https://anr.fr/Project-ANR-19-CE20-0024
<i>Partners</i>	MIA Paris-Saclay, AGAP, LEPSE (INRAE)
<i>Involvement</i>	Team leader, 60,000 € for MIA-Paris
<i>Description</i>	G2WAS aims to identify the genetic regulators of grapevine adaptation to water deficit through genome-wide association studies (GWAS), to support the breeding of drought-tolerant varieties in the context of climate change.
<i>Support</i>	French National Research Agency (ANR)
2019–2024	SINGLESTATOMICS – http://anr-singlestatomics.pages.math.cnrs.fr
<i>Partners</i>	University of Lyon 1, Mines ParisTech, ENS Lyon, AgroParisTech/INRA
<i>Involvement</i>	12 month, co-PI, Team leader , 210,000 € for MIA-Paris
<i>Description</i>	SingleStatOmics develops statistical and machine learning methods for single-cell genomics, to study cell identity, cell differentiation dynamics, gene regulation and single-cell epigenomics, combining expertise in statistics, machine learning and optimal transport.
<i>Support</i>	French National Research Agency (ANR)

2023–2024	BEHIND THE COUNT'HER – Estimation de paramètres génétiques pour des données de comptage en grande dimension à l'aide du modèle Poisson-lognormal
<i>Partners</i>	INRAE (GenPhyse, MIA-PS, MaIAGE), Sorbonne U
<i>Support</i>	Metaprogramm DIGIT-BIO (Meta-omics and microbial ecosystems, INRAE)
2019–2023	ECONET – ADVANCED STATISTICAL MODELLING OF ECOLOGICAL NETWORKS – https://cmatias.perso.math.cnrs.fr/ANR_EcoNet.html
<i>Partners</i>	Sorbonne, Lyon 1 and Lille Universities, AgroParisTech/INRAE, ISEM, IEES
<i>Description</i>	EcoNet develops statistical methods for the analysis of ecological networks (trophic, mutualistic, competitive, host-parasite), integrating space and time, multiple interaction types, covariate information and sampling effects, to predict how ecosystems respond to environmental change.
<i>Support</i>	French National Research Agency (ANR)
2022–2023	PEERSIM – Caractérisation de stress multiple chez les plantes
<i>Partners</i>	INRAE (IPS2, MIA-PS, MIAT), IMT
<i>Support</i>	Metaprogramm DIGIT-BIO (Meta-omics and microbial ecosystems, INRAE)
2022–2023	GENIALEARN – Intérêts et limites de l'apprentissage statistique et de l'apprentissage profond appliqués à la sélection génomique multi-caractères
<i>Partners</i>	INRAE (GABI, MIA-PS), UEVE
<i>Support</i>	Metaprogramm DIGIT-BIO (Meta-omics and microbial ecosystems, INRAE)
2022–2023	BOVMovie2Pred – Développement précoce bovin et viabilité: Exploration et Prédiction
<i>Partners</i>	INRAE (MaIAGE, MIA-PS, BREED)
<i>Support</i>	Metaprogramm DIGIT-BIO (Meta-omics and microbial ecosystems, INRAE)
2018–2022	NEXT-GEN. BIOMONITORING OF CHANGE IN ECOSYSTEMS STRUCTURE AND FUNCTION
<i>Partners</i>	AgroParisTech, INRAE (Dijon, Bordeaux, Rennes, Réunion), Imperial College, Cirad
<i>Description</i>	Next Generation Biomonitoring combines high-throughput sequencing with statistical network reconstruction to detect ecosystem change across six ecosystems (microbial to macro-invertebrate), covering drivers such as invasion, disease, conservation and climate change.
<i>Support</i>	French National Research Agency (ANR)
2018–2019	KINETICKS – Network and modelling analyses to describe the dynamics of Ixodes ricinus microbiome and its influence in pathogen evolution
<i>Partners</i>	INRAE (BIPAR, MaIAGE, MIAP)
<i>Support</i>	Metaprogramm MEM (Meta-omics and microbial ecosystems, INRA)
2017–2019	SEARS – STRATÉGIES D'ÉCHANTILLONNAGE ET ANALYSE DES RÉSEAUX D'APPROVISIONNEMENT EN SEMENCES
<i>Leader</i>	Mathieu Thomas, (CR Cirad, AGAP)
<i>Involvement</i>	1 month
<i>Support</i>	MP GloFoods
2016–2018	BRASSICADIV-PATHO: Microbial diversity and microbial networks associated to Brassica napus and its pathogens
<i>Leader</i>	Christophe Mougél (DR Inra, Rennes), Thierry Candresse (DR Inra, Bordeaux)
<i>Partners</i>	UMR IGEPP, UMR BFP, UMR BioGeCo, UMR EPGV, UMR BioGer
<i>Support</i>	Metaprogramm MEM (Meta-omics and microbial ecosystems, Inra)
2016–2018	LIONS – Large-scale Integrative approach to unravel the complex relationships between differentiation and tumorigenesis
<i>Partners</i>	IGMM/IBC, MAP5, iSSB Évry, Institut Curie, University of York
<i>Support</i>	Plan Cancer 2015 Inserm
2016–2018	LEARNBIOCONTROL: LEARNING ECOLOGICAL NETWORKS FROM METABARCODING DATA: APPLICATION TO BIOLOGICAL CONTROL
<i>Partners</i>	INRA/UMR BioGeCo, Imperial College, AgroParisTech/INRA
<i>Support</i>	MP MEM (Inra)
2015–2018	HYDROGEN – Comparative Metagenomic for Measuring Biodiversity
<i>Partners</i>	AgroParisTech/INRA, CEA-CNS-LABIS, INRIA Rennes/Genscale
<i>Support</i>	French National Research Agency (ANR)

2012–2016	ABS4NGS – Algorithmic, Bioinformatic and Software solutions for the analysis of Next Generation Sequencing data
<i>Partners</i>	Institut Curie, Mines ParisTech, University of Lyon 1, AgroParisTech/INRA
<i>Support</i>	Investissement d’avenir
2014–2016	AREA – Analyse de la Réponse Evolutive des Arbres forestiers tropicaux dans l’environnement, approche génomique et métabolomique
<i>Partners</i>	AgroParisTech/INRA, UMR EcoFoG, UMR 8638 (CNRS/P5)
<i>Support</i>	Défi CNRS “Enviromics”
2015–2016	BEFAST – Deriving Better learning procedures from FASTer algorithms to deal with a huge amount of Data
<i>Leader</i>	Alain Célisse, Assoc. Prof., University Lille 1
<i>Support</i>	PEPS CNRS Fascido
2013–2015	REG4SEL – Regularized methods for Genomic Selection
<i>Leader</i>	Tristan Mary-Huard, CR INRA/AgroParisTech
<i>Partners</i>	UMR MIAP, UMR Le Moulon, GABI
<i>Support</i>	SelGen/French National Institute for Agricultural Research (INRAE)
2011–2015	PLOID-PLOID WHEAT – Unraveling bases of polyploidy and aneuploidy responses in flowering plants, using the wheat ploid model
<i>Partners</i>	INRA (Rennes, Versailles, Grignon), Génoscope, CNRS
<i>Support</i>	French National Research Agency (ANR)
2013–2014	ENORM – Enumeration of Near-Optimal Regulation Misbehaviours
<i>Leader</i>	Étienne Birmelé, PR University Paris 5
<i>Support</i>	PEPS CNRS
2009–2011	NEMO – Network Motif in Biological Network
<i>Partners</i>	AgroParisTech/INRA, University of Lyon 1, University of Évry
<i>Support</i>	French National Research Agency (ANR)
2005–2008	GD2GS – From Genomic Data to Graph Structure
<i>Leader</i>	Florence d’Alché-Buc, Prof. Évry
<i>Support</i>	French National Research Agency (ANR)

STUDENTS

PHD AND POST-DOC - CURRENT

from 2024	BLANCHE FRANCHETERRE
<i>PhD</i>	<i>Interpretable Methods in Unsupervised and Supervised Learning for Multisource analysis of Exposome Data and Prediction of cancer outcomes</i>
<i>Supervision</i>	supervision (50%) with M. Chadeau, Senior Researcher, Imperial College
<i>Support</i>	Horizon Europe (DISCERN)
<i>Summary</i>	Blanche’s PhD develops high-dimensional variable selection and network inference methods for proteomic cancer data, as part of Work Package 5 of the European DISCERN project. Her first paper, from her Master’s research, applied high-dimensional variable selection to plasma proteomic data to identify biomarkers distinguishing stages of human papillomavirus (HPV)-related cervical carcinoma. Building on this applied work, her thesis now focuses on comparing molecular networks across conditions rather than single biomarkers: she introduced Data Shared Neighborhood Selection (DSNS), a joint graphical model that decomposes each condition’s network into a shared component and condition-specific perturbations, extending the Data Shared Lasso to neighborhood selection. She is currently pursuing the development of additional models that attempt to disentangle adverse effects, in the form of latent factors, from variables directly associated with a specific cancer subtype.

from 2024	FRANÇOIS VICTOR
PhD	<i>Développement de méthodes d'apprentissage statistique et d'apprentissage par transfert pour la caractérisation du potentiel agro-écologique et le screening des ressources génétiques</i>
Supervision	supervision (33%) with T. Mary-Huard, DR INRAE and J.B. Léger (MCU Compiègne)
Support	PEPR Agroécologie Numérique (CoBreeding)
Summary	François's PhD develops statistical and deep learning methods to predict the performance of maize hybrids that can be formed from pools of parental lines, as part of the CoBreeding project (PEPR Agroécologie Numérique). Because only a small, unbalanced fraction of possible hybrid combinations is ever observed in the field, the work combines state-of-the-art mixed models from quantitative genetics with neural network architectures (variational autoencoders) to scale inference to large genomic and phenotypic datasets while preserving interpretability. The thesis addresses three complementary questions: predicting the stability of a hybrid's performance across environments from a generalized mixed model with genotype-dependent residual variance, predicting hybrid performance directly in target environments by reformulating factor-analytic and AMMI genotype-by-environment models in a variational framework, and transferring information learned from a data-rich reference population (US maize lines, from the Genomes To Fields initiative) to a more limited population of European lines through transfer learning and domain adaptation.
2023 – 2026	SIHAN XIE
PhD	<i>DeepSelectGene: Apprentissage profond à partir de données de génotypes et application à la sélection génomique</i>
Supervision	supervision (33%) with E. Barrey, DR INRAE and B. Hanczar (PR Évry)
Support	Metaprogramme DigitBio (INRAE)
Summary	Deep learning models are increasingly used to predict phenotypes from genotypes, in human health as in genomic selection, but training them requires many genotype-phenotype pairs, rarely available for species genotyped on only a few thousand animals. Sihan's PhD works around this limit by chaining two models: a generative model, trained on scarce but representative real data, simulates additional genotypes (50–800K SNPs), and a simpler model dedicated to prediction is then trained on this augmented dataset, building on the exploratory work of the GenIALearn project on bovine genomic selection. His first paper compares several families of generative models (VAEs, diffusion models, GANs) for simulating discrete genotypes and shows that they reproduce genetic patterns and preserve genotype-phenotype associations, which also opens the way to privacy-preserving data sharing. A second line of work addresses representation learning for genotypes: since the self-supervised objectives borrowed from text and images carry no biological meaning for a marker profile, the proposal is to borrow that objective from quantitative genetics itself, using kinship between individuals as a signal independent of any phenotype, so that the learned representation's geometry reflects genetic relatedness before any trait is observed.

2023 - 2026	JEANNE TOUS
PhD	<i>Graphical and latent variable models for species community modelling in ecology.</i>
Supervision	100%
Support	PFI (IPEF)
Summary	Deciphering the processes that govern the distribution of species has long been a major goal of ecology. It is also critical to predict biodiversity responses to future changes and devise reliable conservation strategies. These processes (dispersal, abiotic factors, biotic conditions) can usually not be directly observed and must instead be inferred from observational data, mainly species community distribution data and environmental descriptors, but their complexity and their intermingled influences make them hard to decipher. Among the most recent community ecology methods designed to study this question, Joint Species Distribution Models (JSDMs) are a statistical approach, usually based on multivariate latent variable models, that simultaneously model the distributions of all the species in a community, in relationship with the effects of environmental covariates, and infer a network of inter-species residual (marginal or partial) correlations. In this thesis, we develop original models that mix graphical models (for association networks) with multivariate latent variable models (to figure distribution data) while integrating different types of variables clustering. We show that the modelling of zero-inflation in community data can be critical to the JSDM-based analysis of species association networks, and we study the relationship between species competition/facilitation interactions and JSDM-inferred correlations.

PHD AND POST-DOC - ALUMNI

2021 - 2024	BASTIEN BATARDIÈRE
PhD	<i>Machine learning for multivariate analysis of high-dimensional count data</i>
Supervision	supervision (50%) with J. Kwon (50%), CR INRAE
Now	Developer / Data Scientist, Nestlé
Support	ANR SingleStatOmics
Summary	This thesis develops statistical and computational tools for the analysis of high-dimensional multivariate count data, within the framework of the Poisson Log-Normal (PLN) model and its variational inference. It introduces a Zero-Inflated PLN (ZIPLN) extension to account for excess zeros frequently observed in real count data (e.g., microbiome or single-cell data), an adaptive optimization method (AdaLVR) that improves the scalability of variational inference to datasets with several thousand variables, and corrected variational estimators for quantifying parameter uncertainty. The resulting methods are implemented in the PLNmodels and pyPLNmodels R/Python packages and applied to genomic, ecological, and microbiome count datasets.
2017-2020	MARTINA SUNDQVIST
PhD	<i>Multi-omic data integration for new classification in triple-negative breast cancer</i>
Supervision	50% with T. Dubois DR, Institut Curie and G. Rigauil, CR, INRA
Now	Research Engineer, Inserm, Gustave Roussy
Support	DIM Île-de-France
Summary	This thesis addresses the problem of selecting the number of groups in unsupervised clustering and assessing the stability of the resulting partitions, with an application to the molecular classification of triple-negative breast cancers from proteomic data. It introduces stability-based criteria to choose a robust number of clusters and proposes the Multinomial Adjusted Rand Index (MARI), a corrected version of the classical Adjusted Rand Index that removes an unrealistic hypergeometric assumption on cluster sizes, giving a more accurate measure of agreement between clusterings. These tools are used to build a robust, reproducible subtyping of triple-negative breast cancer patients.

2017–2020	AUDREY HULOT
PhD	<i>Analyse de données-omiques: clustering et inférence de réseaux</i>
Supervision	33% with F. Jaffrezic, DR, Inra (33%); H.-J. Garchon, PUPH, Inserm (33%)
Now	Research Engineer INRAE, IJPB
Support	Co-financement MathNum-INRAE / Inserm
Summary	This thesis develops statistical methods for the analysis of high-dimensional omics data, focusing on two complementary tasks: consensus clustering and network inference. It introduces a fast tree aggregation method that combines several hierarchical clustering trees, obtained for instance from different data types or resampled datasets, into a single consensus tree, with an efficient algorithm implemented in the mergeTrees R package. The thesis also contributes methods for inferring biological association networks from omics data, and illustrates both lines of work on real transcriptomic and proteomic datasets.
2016–2019	TIMOTHÉE TABOUY
PhD	<i>Modeling and inferring sampling design in probabilistic random network models</i>
Supervision	50% with P. Barbillon, Assoc. Prof., AgroParisTech
Now	General practice (medicine)
Support	FMJH IDI
Summary	This thesis studies the Stochastic Block Model (SBM) for network data collected under an imperfect, partially observed sampling design. Building on a variational inference framework, it develops estimators that explicitly account for the sampling process, so that latent block structure can be recovered without inferring artefactual clusters induced by missing pairs of observations. The thesis introduces a generic characterization of sampling designs and studies their identifiability with respect to the SBM parameters, and provides the missSBM R package implementing the resulting methodology. Applications are given to seed exchange networks and ecological interaction networks, where sampling is typically incomplete and non-random.
2016–2019	MARIE PERROT-DOCKES
PhD	<i>Regularization tools for multivariate analysis: application to multi-omics</i>
Supervision	50% with Céline Lévy-Leduc, Prof., AgroParisTech
Now	Assoc. Prof., MAP5, Université Paris Cité
Support	FMJH
Summary	This thesis develops regularized methods for variable selection in high-dimensional multivariate linear models, where the response is a vector of correlated outcomes rather than a single variable. It proposes procedures that jointly estimate the regression coefficients and the covariance structure of the residuals, improving selection accuracy when responses are strongly dependent, and establishes theoretical guarantees for the resulting estimators. The methodology is implemented in the MultiVarSel R package and applied to metabolomics data (LC-MS), where it is used to identify which molecular features are jointly associated with experimental conditions while accounting for the complex correlation structure between metabolites.
2013-2016	TRUNG HA
PhD	<i>Statistical learning and multivariate analysis for robust regulatory network inference</i>
Supervision	25% with M.-L. Martin, DR INRA/URGV and G. Rigai, Assoc. Prof., Évry
Now	Biostatistician, Vietnam
Summary	This thesis develops a penalized multivariate learning approach that jointly infers gene expression levels and the underlying gene regulatory network, rather than treating network inference as a separate step performed on already-estimated expression data. By coupling the two estimation problems within a single penalized framework, the approach aims to improve the accuracy of both the expression estimates and the inferred regulatory relationships, particularly when expression data are noisy or partially observed.
2015	DAVID BAKER
Post-doc	<i>Regularization methods for genomic selection</i>
Supervision	50% with Tristan Mary-Huard, CR INRA/Moulon
Now	Statistical Engineer, Inserm
Support	SelGen (Reg4Sel)

2011-2014 <i>PhD</i>	SMAHANE CHALABI <i>Caractérisation de la reprogrammation de l'expression des gènes induite par l'allopolyploidie chez le blé</i>
<i>Supervision</i>	25% with Boulos Chaloub, DR INRA/URGV, Évry
<i>Now</i>	Researcher and Engineer, GenHotel, Université d'Évry-Paris-Saclay
<i>Support</i>	ANR Ploid-Ploid wheat
<i>Summary</i>	This thesis investigates how gene expression is reprogrammed following allopolyploidization in wheat, the process by which a new species arises from the combination and duplication of two parental genomes. Using transcriptomic data from newly formed and stabilized wheat allopolyploids, it characterizes patterns of non-additive gene expression – genes whose expression in the hybrid departs from the simple average of the parental levels – and relates these patterns to genome-wide gene expression additivity previously observed in genetically stable wheat allohexaploids.
2012-2013 <i>Post-doc</i>	ÉDITH LE FLOCH <i>Analysis of NGS data to characterize polyploidy</i>
<i>Supervision</i>	50% with Carène Rizzon, Assoc. Prof., Évry
<i>Now</i>	Research Engineer, CEA
2011-2013 <i>PhD</i>	JONATHAN PLASSAIS <i>Développement méthodologique pour la méta-analyse appliquée à la caractérisation de signatures chez les patients atteints de maladie auto-immune</i>
<i>Supervision</i>	50% with Christophe Ambroise, Prof., Évry
<i>Now</i>	Senior Biostatistician, ScientaLab
<i>Support</i>	CIFRE, société TcLand www.tcland-expression.com
<i>Summary</i>	This CIFRE thesis, carried out with the company TcLand Expression, develops statistical methodology for the meta-analysis of transcriptomic studies, with the aim of characterising molecular signatures in patients with autoimmune diseases. Combining studies conducted on different platforms, protocols and patient populations raises specific difficulties, from between-study heterogeneity to the multiplicity of tests, which the work addresses by adapting variable selection and aggregation procedures to this setting so that the signatures obtained are reproducible from one cohort to another rather than specific to a single study.
2009–2012 <i>PhD</i>	CAMILLE CHARBONNIER <i>Inference of gene regulatory networks from non-iid transcriptomic data</i>
<i>Supervision</i>	50% with Christophe Ambroise, Prof., Évry
<i>Now</i>	Assoc. Prof. / Hospital Practitioner, Université de Rouen / CHU de Rouen
<i>Support</i>	EDGO Évry
<i>Summary</i>	This thesis develops statistical methods for inferring gene regulatory networks from transcriptomic data that violate the classical independent and identically distributed (iid) sampling assumption, in particular time-course and structured experimental designs. It proposes weighted-Lasso and cooperative-Lasso penalized estimators of Gaussian graphical models that incorporate prior structural information (temporal ordering, replicate structure) to improve network recovery, together with model selection procedures adapted to these dependent settings. The resulting methods are implemented in the SIMoNe R package and applied to real transcriptomic time-course data.

MASTERS - ALUMNI

2026 <i>Supervision</i>	KATERYNA STETSUN (6 months), Université Paris-Saclay with G. Rigai, A. Leroy
<i>Master</i>	<i>Détection de rupture multivariée et classification de courbes pour l'analyse de cinétique de métabolites</i>
2026 <i>Supervision</i>	NESTOR NGALALA MANGUITINI (6 months), Université Paris-Saclay with J. Tous
<i>Master</i>	<i>Un modèle lognormal multivarié inflaté en zéro avec clustering dans les effets fixes</i>
2024 <i>Supervision</i>	ALEXANDRE PRADIGNAC (6 months), Université Paris-Saclay with T. Mary-Huard, L. Sansonnet
<i>Master</i>	<i>Modélisation statistique pour la dynamique physiologique des plantes</i>

2023 <i>Supervision</i> <i>Master</i>	CAMILLE MONDON (6 months), ENS Ulm with M. Mariadassou (MaIAGE, INRAE) <i>Poisson log-normal estimation with dependency between sites</i>
2023 <i>Supervision</i> <i>Master</i>	FRANÇOIS VICTOR (6 months), Univ. Claude Bernard Lyon-1 with J. Kwon, T. Mary-Huard, J.-B. Léger <i>Prédiction du phénotype à partir du génotype par apprentissage automatique</i>
2023 <i>Supervision</i> <i>Master 1</i>	PIERRE BRAND (3 months), Univ. Paris Saclay, MsC. "Mathématiques Fondamentales" with P. Barbillon <i>ACP probabiliste et généralisation</i>
2023 <i>Supervision</i> <i>Master 1</i>	KHALED BOUGILA (5 months), ENSTA with Nicolas Jouvin (80%) <i>Caractérisation de stress multiple chez les plantes par analyse conjointe de données omiques à l'aide de modèles Poisson lognormaux</i>
2022 <i>Supervision</i> <i>Master</i>	JIANSU ZHU (6 months), Télécom Paris with J. Kwon, S. Bord <i>Prédiction multivariée de 35 caractères de production du lait par gradient boosting</i>
2021 <i>Supervision</i> <i>Master</i>	ARMAND FAVREAU (6 months), Univ. Paris Saclay, MsC. "Math et Science du Vivant" with S. Donnet, J. Aubert and I. Albert (INRAE/AgroParisTech) <i>Analyse en composantes principales structurée et parcimonieuse</i>
2021 <i>Supervision</i> <i>Master</i>	BASTIEN BATARDIÈRE (6 months), University of Dauphine/MVA with J. Kwon, L. Sansonnet (INRAE/AgroParisTech) <i>Modèles de Poisson lognormal multivariés: optimisation, inférence et applications en grande dimension</i>
2021 <i>Supervision</i> <i>Master</i>	HUGUES VAN ASSEL (6 months), École Polytechnique/MVA with F. Picard (DR CNRS, École centrale de Lyon), T. Espinasse (MCF University of Lyon 1) <i>Probabilistic models for graph Embedding</i>
2019 <i>Supervision</i> <i>Master</i>	CLAIRE GAYRAL (6 months), University of Lyon 1 with F. Picard (DR CNRS, LBBE, Lyon) <i>Integrating epigenetic and expression data for subclone discrimination in single-cell</i>
2017 <i>Supervision</i> <i>Master 1</i>	RÉMI BERNHARD (2 months), ENSIIE with T. Flûtre, T. Mary-Huard CR INRA, L. Sansonnet, AgroParisTech <i>Structured variable selection for multi-trait GWAS in grapevine</i>
2017 <i>Supervision</i> <i>Master</i>	MARTINA SUNDQVIST (6 months), ENS, Paris-Descartes with T. Dubois, L. De Koning, Institut Curie; G. Rigai, CR Inra/BAP <i>Clustering for proteomic and transcriptomic analysis of basal breast cancer</i>
2016 <i>Supervision</i> <i>Master</i>	AUDREY HULOT (6 months), ENSAI with G. Rigai, CR and F. Jaffrezic, DR INRA <i>Clustering convexe à large échelle pour la métagénomique</i>
2016 <i>Supervision</i> <i>Master</i>	TIMOTHÉE TABOUY (6 months), Master Math et Science du Vivant, Paris-Saclay with P. Barbillon, S. Ouadah, Assoc. Prof., AgroParisTech; S. Donnet, CR Inra <i>Modeling and inferring sampling design in probabilistic random network models</i>
2016 <i>Supervision</i> <i>Master</i>	MARGOT BRÉGÈRE (6 months), Master Math et Science du Vivant, Paris-Saclay 33% with C. Lévy-Leduc, Prof. and L. Sansonnet, Assoc. Prof., AgroParisTech <i>Variable selection in Multivariate ANOVA for ecological data</i>
2015 <i>Supervision</i> <i>Master</i>	VALENTIN DERVIEUX (6 months), Télécom Sud Paris 50% with Guillem Rigai, Assoc. Prof., Évry <i>Clustering et analyse multivariée de données métagénomique du porc</i>
2012-2013 <i>Supervision</i> <i>Master</i>	PIERRE GUTIERREZ (6 months + 3 months CDD), École Nationale de la Statistique et de l'Administration (ENSAE) 50% with Guillem Rigai, Assoc. Prof., Évry <i>Multi-class differential analysis with fused-Anova</i>

2011 <i>Supervision</i> <i>Master</i>	GEN YANG (3 months), ENSIIE 50% with Christophe Ambroise, Prof., Évry <i>Hierarchical Lasso and group-Lasso for gene selection</i>
2011 <i>Supervision</i> <i>Master</i>	AUORE MOUTARDE (5 months), MIGS, Université de Bourgogne 50% with Yves Grandvalet, DR, UTC <i>Développements algorithmiques dans les méthodes de régression pénalisée appliquées à la sélection de gènes</i>
2010 <i>Master</i>	CYRILLE LONGIN (6 months), EGOIS, Université de Rouen <i>Caractérisation automatique de modules fonctionnels dans les réseaux de régulation</i>
2009 <i>Supervision</i> <i>Master</i>	CAMILLE CHARBONNIER (5 months), ENSAE 50% with Christophe Ambroise, Prof., Évry <i>ℓ_1 penalization and application to the inference of sparse dynamic regulation networks</i>
2008 <i>Supervision</i> <i>Master</i>	ALEXANDER SMITH (6 months), AgroCampusOuest 50% with Christophe Ambroise, Prof., Évry <i>Développement d'une nouvelle méthode d'estimation de réseaux de régulation</i>

TEACHING ACTIVITIES

Approximately 2000 hours of undergraduate and Master's teaching in applied mathematics, statistics, biology and computer science at Université d'Évry, AgroParisTech, École Nationale Supérieure d'Informatique pour l'Industrie et l'Entreprise (ENSIIE), École Nationale de la Statistique et de l'Analyse de l'Information (ENSAI), Université de Technologie de Compiègne, Université Paris-Sud, Université Paris Dauphine and X – École Polytechnique, including as lecturer and teaching supervisor at École Polytechnique (2020–2024) and ENSAI (2015–2018).

Also teach continuing education courses in data science and machine learning for industry professionals (X-Executive Education, Dauphine Executive Education, HEC Data Science Certificate).

I mention here only a few lessons that I consider noteworthy, as they correspond to the production of support material and responsibility for a course.

2020-24 <i>Msc</i> <i>web</i>	STATISTICS IN ACTION WITH R Probabilistic models, data analysis, R programming https://jchiquet.github.io/map566
2020-21 <i>Msc</i> <i>web</i>	DATA ANALYSIS AND UNSUPERVISED LEARNING Data and Graph Clustering, Mixture model, Stochastic Block Model https://jchiquet.github.io/map573
2018-19 <i>Msc</i> <i>web</i>	AN INTRODUCTION TO GRAPH ANALYSIS AND MODELING (36h course/practicals) Descriptive Analysis of networks, Stochastic Block Model, Graphical Lasso https://github.com/jchiquet/CourseStatNetwork
2015-18 <i>Msc</i> <i>web</i>	INTRODUCTION TO REGULARIZATION FOR REGRESSION (154h course/practicals) Ridge, Lasso, variable selection, model selection https://github.com/jchiquet/CourseRegLinearRegression

PAPERS

PREPRINTS

- [PP1] D. Agyapong, J. Chiquet, J. Marks, and T. D. Hocking. *Understanding When Poisson Log-Normal Models Outperform Penalized Poisson Regression for Microbiome Count Data*. 2026. DOI: [10.48550/arXiv.2604.03853](https://doi.org/10.48550/arXiv.2604.03853).
- [PP2] B. Francheterre, R. Colindres Zuehlke, V. Viallon, M. Chadeau-Hyam, and J. Chiquet. *Data Shared Neighbourhood Selection for multi-condition network inference*. 2026. DOI: [10.48550/arXiv.2608.22901](https://doi.org/10.48550/arXiv.2608.22901).
- [PP3] J. Tous and J. Chiquet. *Catching the effects of biotic interactions on community data: partial correlations outperform marginal ones with proper abiotic modelling*. 2026. DOI: [10.64898/2026.05.20.726512](https://doi.org/10.64898/2026.05.20.726512).
- [PP4] J. Tous, J. Chiquet, A. E. Deacon, A. Fontrodona-Eslava, D. F. Fraser, and A. E. Magurran. *A JSMD with zero-inflation to increase the ecological relevance of analyses of species distribution data*. 2025. DOI: [10.1101/2025.07.24.666553](https://doi.org/10.1101/2025.07.24.666553).
- [PP5] B. Batardière, J. Chiquet, and M. Mariadassou. *Evaluating Parameter Uncertainty in the Poisson Lognormal Model with Corrected Variational Estimators*. 2024. DOI: [10.48550/arXiv.2411.08524](https://doi.org/10.48550/arXiv.2411.08524).
- [PP6] B. Batardière, J. Chiquet, and J. Kwon. *Finite-sum optimization: Adaptivity to smoothness and loopless variance reduction*. 2023. DOI: [10.48550/arXiv.2307.12615](https://doi.org/10.48550/arXiv.2307.12615).

JOURNAL PAPERS

- [JP1] R. Hornung, L. Németh, O. Zadorozhnyi, T. Ullmann, M. Kammer, R. Killick, C. Paciorek, J. Chiquet, M. Herrmann, L. Batinović, R. Carlsson, P. Neuvial, B. Hejblum, J. Wrobel, A.-L. Boulesteix, and K. Tabelow. “Overcoming barriers to computational reproducibility”. In: *Royal Society Open Science* 13.6 (2026), p. 252489. DOI: [10.1098/rsos.252489](https://doi.org/10.1098/rsos.252489).
- [JP2] J. Tous and J. Chiquet. “An integrated method for clustering and association network inference”. In: *Computational Statistics & Data Analysis* 219 (2026), p. 108347. DOI: <https://doi.org/10.1016/j.csda.2026.108347>.
- [JP3] S. Xie, T. Tribout, D. Boichard, B. Hanczar, J. Chiquet, and E. Barrey. “Learning inherent genetic patterns and trait associations with deep generative models for discrete genotype simulation”. In: *GigaScience* (2026), giag044. DOI: [10.1093/gigascience/giag044](https://doi.org/10.1093/gigascience/giag044).
- [JP4] B. Batardière, J. Chiquet, F. Gindraud, and M. Mariadassou. “Zero-inflation in the Multivariate Poisson Lognormal Family”. In: *Statistics and Computing* 35.196 (2025). DOI: [10.1007/s11222-025-10729-0](https://doi.org/10.1007/s11222-025-10729-0).
- [JP5] B. Batardière, J. Chiquet, J. Kwon, and J. Stoehr. “Importance sampling-based gradient method for dimension reduction in Poisson log-normal model”. In: *Electronic Journal of Statistics* 19.1 (2025). DOI: [10.1214/25-EJS2384](https://doi.org/10.1214/25-EJS2384).
- [JP6] B. Bodinier, S. Rodrigues, M. Karimi, S. Filippi, J. Chiquet, and M. Chadeau-Hyam. “Stability Selection and Consensus Clustering in R: The R Package sharp”. In: *Journal of Statistical Software* 112.5 (2025). DOI: [10.18637/jss.v112.i05](https://doi.org/10.18637/jss.v112.i05).
- [JP7] B. Batardière, J. Kwon, and J. Chiquet. “pyPLNmodels: A Python package to analyze multivariate high-dimensional count data”. In: *Journal of Open Source Software* 9.104 (2024), p. 6969. DOI: [10.21105/joss.06969](https://doi.org/10.21105/joss.06969).
- [JP8] R. Wada, F.-J. Peng, C.-A. Lin, R. Vermeulen, A. Iglesias-González, P. Palazzi, B. Bodinier, S. Streel, M. Guillaume, D. Vuckovic, et al. “Hair-Derived Exposome Exploration of Cardiometabolic Health: Piloting a Bayesian Multitrait Variable Selection Approach”. In: *Environmental Science & Technology* (2024). DOI: <https://doi.org/10.1021/acs.est.3c08739>.
- [JP9] B. Bodinier, S. Filippi, T. H. Nøst, J. Chiquet, and M. Chadeau-Hyam. “Automated calibration for stability selection in penalised regression and graphical models”. In: *Journal of the Royal Statistical Society Series C: Applied Statistics* (2023). DOI: [10.1093/jrsssc/qlad058](https://doi.org/10.1093/jrsssc/qlad058).
- [JP10] B. Bodinier, S. Rodrigues, M. Karimi, S. Filippi, J. Chiquet, and M. Chadeau-Hyam. “Automated calibration of consensus weighted distance-based clustering approaches using sharp”. In: *Bioinformatics* (2023). DOI: <https://doi.org/10.1093/bioinformatics/btad635>.
- [JP11] E. Delannoy, B. Batardière, S. Pateyron, L. Soubigou-Taconnat, J. Chiquet, J. Colcombet, and J. Lang. “Cell specialization and coordination in Arabidopsis leaves upon pathogenic attack revealed by scRNA-seq”. In: *Plant Communications* 4.5 (2023). Focus Issue on Plant Single-Cell Biology, p. 100676. DOI: <https://doi.org/10.1016/j.xplc.2023.100676>.

- [JP12] H. V. Assel, T. Espinasse, J. Chiquet, and F. Picard. “A Probabilistic Graph Coupling View of Dimension Reduction”. In: *Advances in Neural Information Processing Systems*. Ed. by A. H. Oh, A. Agarwal, D. Belgrave, and K. Cho. 2022. URL: <https://openreview.net/forum?id=tNXumks8yHv>.
- [JP13] P. Barbillon, J. Chiquet, and T. Tabouy. “missSBM: An R Package for Handling Missing Values in the Stochastic Block Model”. In: *Journal of Statistical Software* 101.12 (2022), pp. 1–32. DOI: [10.18637/jss.v101.i12](https://doi.org/10.18637/jss.v101.i12).
- [JP14] M. Sundqvist, J. Chiquet, and G. Rigai. “Adjusting the adjusted Rand Index - A multinomial story”. In: *Computational Statistics* (2022). DOI: [10.1007/s00180-022-01230-7](https://doi.org/10.1007/s00180-022-01230-7).
- [JP15] C. Brault, A. Doligez, L. Le Cunff, A. Coupel-Ledru, T. Simonneau, J. Chiquet, P. This, and T. Flutre. “Harnessing multivariate, penalized regression methods for genomic prediction and QTL detection to cope with climate change affecting grapevine”. In: *G3: Genes/Genomes/Genetics* (2021). DOI: [10.1093/g3journal/jkab248](https://doi.org/10.1093/g3journal/jkab248).
- [JP16] M. Champion, J. Chiquet, P. Neuvial, M. Elati, F. Radványi, and E. Birmelé. “Identification of deregulation mechanisms specific to cancer subtypes”. In: *Journal of Bioinformatics and Computational Biology* 19.01 (2021), p. 2140003. DOI: [10.1142/S0219720021400035](https://doi.org/10.1142/S0219720021400035).
- [JP17] J. Chiquet, M. Mariadassou, and S. Robin. “The Poisson-Lognormal Model as a Versatile Framework for the Joint Analysis of Species Abundances”. In: *Frontiers in Ecology and Evolution* 9 (2021), p. 188. DOI: [10.3389/fevo.2021.588292](https://doi.org/10.3389/fevo.2021.588292).
- [JP18] M. Dubart, P. Alonso, D. Barroso-Bergada, N. Becker, K. Bethune, D. A. Bohan, C. Boury, M. Cambon, E. Canard, E. Chancerel, J. Chiquet, P. David, N. de Manincor, S. Donnet, A. Duputié, B. Facon, E. Guichoux, T. Le Minh, S. Ortiz-Martínez, L. Piuzeau, A. Sacco-Martret de Préville, M. Plantegenest, C. Poux, V. Ravigné, S. Robin, M. Trillat, C. Vacher, C. Vernière, and F. Massol. “Chapter Nine - Coupling ecological network analysis with high-throughput sequencing-based surveys: Lessons from the next-generation biomonitoring project”. In: *The Future of Agricultural Landscapes, Part III*. Ed. by D. A. Bohan, A. J. Dumbrell, and A. J. Vanbergen. Vol. 65. Advances in Ecological Research. Academic Press, 2021, pp. 367–430. DOI: <https://doi.org/10.1016/bs.aecr.2021.10.007>.
- [JP19] B. Facon, A. Hafsi, M. C. de la Masselière, S. Robin, F. Massol, M. Dubart, J. Chiquet, E. Frago, F. Chiroleu, P.-F. Duyck, and V. Ravigné. “Joint species distributions reveal the combined effects of host plants, abiotic factors and species competition as drivers of species abundances in fruit flies”. In: *Ecological Letters* (2021). DOI: [10.1111/ele.13825](https://doi.org/10.1111/ele.13825).
- [JP20] E. Lejal, J. Chiquet, J. Aubert, S. Robin, A. Estrada-Peña, O. Rue, C. Midoux, M. Mariadassou, X. Bailly, A. Cougoul, P. Gasqui, J. Cosson, K. Chalvet-Monfray, M. Vayssier-Taussat, and T. Pollet. “Temporal patterns in Ixodes ricinus microbial communities: an insight into tick-borne microbe interactions”. In: *Microbiome* 9.153 (2021). DOI: [10.1186/s40168-021-01051-8](https://doi.org/10.1186/s40168-021-01051-8).
- [JP21] F. Guinot, M. Szafranski, J. Chiquet, A. Zancarini, C. Le Signor, C. Mougél, and C. Ambroise. “Fast Computation of Genome-Metagenome interaction effects”. In: *Algorithms for Molecular Biology* (2020). DOI: [10.1186/s13015-020-00173-2](https://doi.org/10.1186/s13015-020-00173-2).
- [JP22] A. Hulot, J. Chiquet, F. Jaffrezic, and G. Rigai. “Fast tree aggregation for consensus hierarchical clustering”. In: *BMC Bioinformatics* (2020). DOI: [10.1186/s12859-020-3453-6](https://doi.org/10.1186/s12859-020-3453-6).
- [JP23] C. Pauvert, T. Fort, A. Calonnec, J. Faivre-d’Arcier, E. Chancerel, M. Massot, J. Chiquet, S. Robin, D. A. Bohan, J. Vallance, and C. Vacher. *Microbial association networks give relevant insights into plant pathobiomes*. 2020. DOI: [10.1101/2020.02.21.958033](https://doi.org/10.1101/2020.02.21.958033).
- [JP24] T. Tabouy, P. Barbillon, and J. Chiquet. “Variational Inference for Stochastic Block Models from Sampled Data”. In: *Journal of the American Statistical Association* 115.529 (2020), pp. 455–466. DOI: [10.1080/01621459.2018.1562934](https://doi.org/10.1080/01621459.2018.1562934).
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- [JP26] M. Grandclaude, M. Perrot-Dockès, C. Trichot, O. Mostafa-Abouzid, W. Abou-Jaoudé, F. Berger, P. Hupé, D. Thieffry, L. Sansonnet, J. Chiquet, C. Lévy-Leduc, and V. Soumelis. “A Quantitative Multivariate Model of Human Dendritic Cell-T Helper Cell Communication”. In: *Cell* 179.2 (2019), pp. 432–447. DOI: [10.1016/j.cell.2019.09.012](https://doi.org/10.1016/j.cell.2019.09.012).
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- [JP28] M. Perrot, C. Lévy-Leduc, J. Chiquet, L. Sansonnet, M. Brégère, M.-P. Étienne, S. Robin, and G. Genta-Gouve. “A multivariate variable selection approach for analyzing LC-MS metabolomics data”. In: *SAGMB* (2018). DOI: [10.1515/sagmb-2017-0077](https://doi.org/10.1515/sagmb-2017-0077).

- [JP29] M. Perrot, C. Lévy-Leduc, L. Sansonnet, and J. Chiquet. “Variable selection in multivariate linear models with high-dimensional covariance matrix estimation”. In: *J. Multivar. Anal.* 166 (2018), pp. 78–97. DOI: [10.1016/j.jmva.2018.02.006](https://doi.org/10.1016/j.jmva.2018.02.006).
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